

Carson Mershon

cpmershon@gmail.com ♦ (207) 751-2864 ♦ [linkedin.com/in/carson-mershon/](https://www.linkedin.com/in/carson-mershon/)

EDUCATION

Wentworth Institute of Technology

2021 – 2025

Bachelor of Science, Computer Engineering

Boston, MA

- **Relevant Courses:** Computer Networks for Engineers, Applied Programming Concepts, Object Oriented Programming, Computer Architecture, Computer Operating Systems, VoIP & Unified Communications.

SKILLS

- **Programming Languages:** C, C++, Java, JavaScript, Python, MATLAB, Assembly Language, Machine Language, Verilog HDL, HTML
- **Hardware/Equipment:** Windows/Mac OS, Ubuntu Linux LTS, CNC Machining, DC Power Supply, Function/Waveform Generator, Digital Multimeter, Oscilloscope, Breadboard
- **Software/Tools:** ServiceNow, MATLAB, Microsoft Office Suite, Adobe Creative Suite, Visual Studio Code, Arduino, SolidWorks, Multisim, Git, Intel Quartus II, ESP-IDF, Docker, FreeRTOS, OpenThread

PROJECTS

Wireless Sensor Network to Detect Power Line Outages

2025

- **Technical Achievements:**
 - Architected and prototyped a capacitive-coupling EMF sensing circuit around ESP32-C6 SoCs. The design included input buffering, op-amp gain stages, and 60 Hz band-pass filters to reliably isolate overhead line emissions. Achieved reliable detection in test scenarios with 100% accuracy.
 - Engineered a custom mesh-network protocol using OpenThread; created lightweight UDP packet formats for outage reporting, integrated gateway interfacing, and optimized routing to ensure reliable multi-hop communication in lossy outdoor environments.
 - Programmed ESP-IDF firmware with FreeRTOS for continuous sampling, Fourier analysis, and UDP packet-based outage reporting with reliable end-to-end data delivery to Google Sheets dashboards via a border router and Google Apps Script.

RECENT WORK EXPERIENCE

Technical and Media Support Analyst

January 2024 – July 2025

Simmons University

Boston, MA

- **Key Responsibilities:**
 - Provided comprehensive technical and media support via multiple channels (in-person, phone, email) for faculty, staff, and students, addressing break-fix issues, service requests, and application installations.
 - Actively contributed to technology projects such as desktop upgrades, deploying new computers and software, upgrading computer labs, and facilitating the enhancement of the technology infrastructure.
 - Consistently earned positive feedback from faculty and staff for effective problem resolution.

Wireless Technologies Tech Support and Retail Sales Associate

May 2022 – September 2022

Marketsource – Allegis Group

Brunswick, ME

- **Key Responsibilities:**
 - Provided quick and effective device troubleshooting support to customers with issues related to their wireless devices or related technologies and network configurations.
 - Consistently surpassing sales targets by 200%+ and earning recognition as a top-performing associate.